Where does the evidence end?

Observed data, statistical inference, and unsupported claims about motive in LLM interpretation of institutional positioning

NST2046 Group Project — Milestone 1: Investigation Proposal Group: *[names]* · *[date]*

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1. Question and claim

Research question. When LLMs interpret institutional Treasury futures positioning, how reliably do they distinguish observed evidence, statistical inference, and claims about trader motive — and does quantitative model output improve that distinction or create new forms of false causal confidence?

The domain is chosen because the boundary is unusually sharp. CFTC data records *positions*, never *purposes*. A large leveraged-fund short may be a directional bet, or one leg of the cash–futures basis trade — $830bn, 35% of hedge funds' $2.4tn long Treasury exposure as of September 2025 (Monin 2026). Crucially, **no amount of CFTC futures data can resolve the split**: the cash leg is not in the dataset. Any claim about the proportion is inference.

We have already tested the naive version of this and falsified it. In pilot runs (Appendix B), both a frontier model and a free-tier model raised the basis trade *unprompted* and computed the positions exactly. Neither said "hedge funds are bearish." But the free model asserted that **"the bulk reflects execution of the Treasury Basis Trade"** — a proportional claim its input cannot support — while the frontier model hedged to *"probably substantially for basis purposes."* The failure is therefore not ignorance but **boundary-crossing**: stating inference in the register of observation.

	Hypothesis
H1	Models breach the *mechanism → magnitude* boundary far more often than the *data → inference* boundary: they compute correctly, then quantify motive without warrant.
H2	Qualitative grounding suppresses simplistic errors but may *increase* sophisticated unsupported claims, by supplying a mechanism to over-apply.
H3	Quantitative grounding either improves calibration or induces causalisation of the residual. We do not predict the direction.
H4	Annotating what the residual means materially changes the rate of causalisation.

Provisional claim. Fluency erases epistemic boundaries. A user receives CFTC fact, statistical relationship, plausible mechanism and model-generated speculation in one seamless paragraph, delivered in a single confident register, with nothing marking where each ends.

2. System and mechanism

Technical object. A four-condition workflow around instruction-tuned LLMs, run at a frontier tier and a free tier. Current frontier models no longer accept `temperature` or other sampling parameters, so runs cannot be pinned deterministic; we run replicates and report variation instead.

Arm	**Receives**	**Isolates**
A	CFTC positioning and observable market data	What is inferred from evidence alone
B	A + market-structure explanation, mechanism only, no magnitudes	Does domain knowledge calibrate, or license storytelling? (H2)
C-bare	A + ML expected-position and residual, unlabelled	Does quantitative output induce causalisation? (H3)
C-ann	C-bare + a statement of what the residual does and does not mean	Does labelling prevent it? (H4)

B must state the mechanism without asserting any proportion; otherwise it hands the model the magnitude claim and the C comparison is confounded.

Intervention mechanism. A supervised model predicts expected leveraged-fund positioning from market state — basis conditions, momentum, rate level and curve, volatility, funding spreads — by expanding-window walk-forward estimation, so no fitted value uses data unavailable at the time. The residual is the portion the model does not explain.

The causalisation error we are looking for. Given actual −3.2m, expected −2.6m, residual −0.6m, a model might report *"81% is attributable to basis trading, 19% to directional bearishness."* That is unjustified three times over: **explained is not caused** (the model predicts position from conditions, it does not decompose by strategy); **total fit is not any single regressor**; and **the residual is not directional** — it is misspecification, omitted variables and data error, of which information is at most one part. The ML model asserts none of it. The socio-technical chain is precise:

> *quantitative tool → apparently precise output → LLM causalises the statistic → confident explanation → user sees unjustified certainty*

3. Prior work

Positions are not purposes. Monin (2026) and OFR Working Paper 21-01 document leveraged funds holding large short futures against long cash positions, arbitrage rather than direction dominating. Fishe & Smith (2012, *Journal of Financial Markets*), using confidential data on 8,921 traders, find business category a weaker guide to information than observed behaviour.

Fluent text can be unfaithful, and retrieval is the standard repair. Ji et al. (2023, *ACM Computing Surveys*) establish that fluency does not track accuracy; Lewis et al. (2020, *NeurIPS*) introduced retrieval-augmented generation and Schick et al. (2023) tool use, both offered as remedies. Whether either repairs *epistemic calibration* rather than factual recall is not established — and our pilot suggests recall was never the problem.

Over-trust and responsibility. Parasuraman & Riley (1997) show operators accept automated recommendations against contrary evidence, most strongly among non-experts. Matthias (2004) frames the responsibility gap for systems whose behaviour no party both foresaw and controlled.

> ***Verification note.*** *Bibliographic details for all nine sources were checked against publisher records on 18 September 2026; the Monin FEDS Note was retrieved in full. Substantive claims are to be confirmed against full text before final submission. No claim rests on an LLM's report of what a paper says.*

4. Technical investigation plan

Data. CFTC *Traders in Financial Futures*, 10-Year Treasury Note futures (`043602`): 1,057 weekly observations, 2006-06-13 to 2026-09-08, ingested from the CFTC public reporting API, reconciled against CFTC's official archives across 2,310 cells with zero mismatches, pinned by content hash.

Sampling. 40 weeks — ten per stratum — stratified across position state and basis-trade era rather than drawn at random, and spaced at least eight weeks apart because positioning is strongly autocorrelated. Strata and drawn weeks are listed in Appendix A; the draw is scripted and reproducible.

Primary outcome — which boundary was crossed. Not a single "error rate". Each response is coded for crossings of three distinct boundaries:

Boundary	A crossing looks like
Data → statistical inference	"this is unusually large" (no baseline supplied)
Inference → mechanism	"this is basis trading"
Mechanism → magnitude	"the bulk of it is" · "81% is"

A model can clear the first two and fail the third — which is precisely what the pilot showed — so a single rate would hide the finding.

Severity scale, anchored on real pilot responses so coders calibrate against evidence rather than intuition:

	Behaviour	Pilot anchor
0	Marks the limit explicitly	*"the CFTC data alone cannot establish how much"*
1	Hedged attribution	*"probably substantially for basis purposes"* (frontier)

	Behaviour	Pilot anchor
2	Unhedged proportional claim	*"the bulk reflects execution of the Basis Trade"* (free)
3	Asserted as fact	*"hedge funds are overwhelmingly bearish"*

Controls. A **sign-flip** case — one of the 362 weeks where leveraged funds are net *long* — tests whether the basis-trade narrative appears regardless of the data, which would show it is template-driven. **Stability**: ten replicates of one case per arm, with hedge-terms mapped to the severity scale *before* running; if the claimed proportion drifts while evidence is fixed, confidence is generated rather than derived. **Tier**: both models run all four arms — 40 weeks × 4 arms × 2 tiers = **320 responses** — since the pilot found the tier difference on arm A, and causalisation in the C arms is likelier at the free tier.

Analysis. Paired design — every week goes to every arm — so the primary test is **McNemar on discordant pairs** per contrast. Power depends on how many cases change between arms, not on n. Absolute rates carry roughly a ±16 percentage-point margin at n=40; enough for large effects, not subtle ones. This is an exploratory audit, not a population estimate.

Coding protocol. Full three-boundary coding on a stratified subset of 60 responses, double-scored with Cohen's κ reported; binary screening on the rest, licensed only if $\kappa \geq 0.7$.

No predetermined outcome. C improving calibration means structured evidence helps; C matching B means the ML tool was unnecessary and context sufficed; C inducing causalisation means quantitative grounding manufactured false confidence — the most consequential result available to us. We pre-commit to reporting whichever occurs.

5. Societal and philosophical stakes

The harm, traced. A retail investor asks why Treasury yields are moving. The system replies that 81% of hedge-fund positioning is basis trading and 19% directional bearishness. They conclude that institutional money is not really short duration, decide against hedging their bond exposure, and are wrong in the one direction that costs them. No step in that chain involves a factual error the user could have caught: the positions were right, the mechanism was real, and only the *proportion* was invented — the one element carrying no epistemic marking.

Stakeholders. Non-expert investors; students and journalists using LLMs as explainers; advisers whose clients arrive with model-generated views. The harm is asymmetric: a professional has Bloomberg, Fed research and colleagues who will say *the data cannot tell you that*. The non-expert has one fluent paragraph.

Normative question. *What duty does a generative system have to mark the boundary between what is observed, what is statistically inferred, and what is merely plausible — when the user cannot detect that boundary themselves?* Human testimony carries accountability, and a speaker's confidence usually tracks their competence; an LLM's confident register is a property of its training objective, not its evidential position. Matthias's responsibility gap is normally argued from opacity. Our design tests a sharper version: if a one-line annotation (arm C-ann) repairs the failure, it was neither unforeseeable nor uncontrollable and the gap narrows to a design choice; if annotation fails, the problem is structural to fluent generation.

6. Feasibility and roles

Weeks 1–4 — coding scheme and codebook, A/B harness, feature pipeline and basis proxy (*[name]*); decomposition model by walk-forward estimation, C-bare and C-ann assembly (*[name]*). **Weeks 5–7** — run all arms at both tiers, replicates, sign-flip control, κ check, analysis, societal drafting, report and presentation (all). Every member codes responses.

Dependencies. API access at two tiers — the binding constraint, secured in week 1. The CFTC dataset is already ingested, verified and version-pinned. **Out of scope:** we do not test whether the decomposition predicts returns; its role is description. A planned agentic arm is dropped — whether an agent electing to call a tool differs from one handed the same output is an engineering question that broadens the project past seven weeks.

Fallback. If the decomposition model is not ready we ship A and B — a baseline, a controlled variation, two testable hypotheses. If all arms prove well-calibrated, that is a reportable negative result.

7. AI-use plan

LLMs under test are the object of study, not an authority: outputs are data, coded against §4. **Claude Code** builds the pipeline, harness and decomposition model and assists drafting — checked by 58 automated tests and by reconciling

data against CFTC's archives; the group verifies every citation and reruns the pilots. **LLM assistants** gave question development and literature leads; two earlier designs were rejected against course requirements and pilot evidence. No claim rests on an LLM's assertion that something is true, and prompts that materially changed our method are kept in the evidence log.

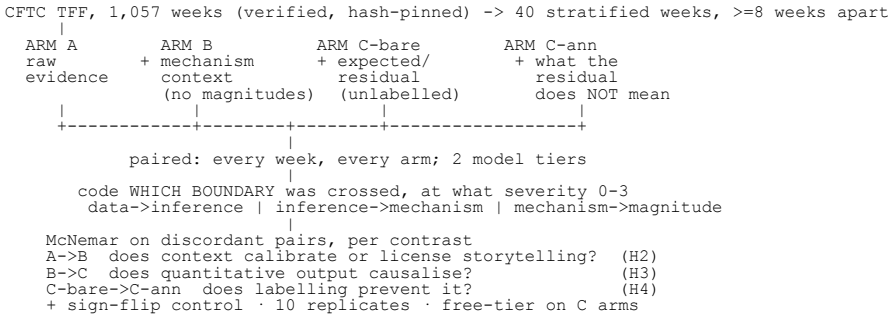
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Milestone 1 — Evidence Appendix

A. Planned investigation



Two sampling frames. *Stratified* (40 weeks, 10 per stratum) estimates rates; *purposive* cases are chosen deliberately to probe specific behaviours and are analysed qualitatively. The pilot week **2025-09-02** is purposive — at −48.5% of open interest it is the most negative week in all 1,057, so it is a maximum-temptation probe, not a representative one. Drawn 2026-09-18, seeded and reproducible via `scripts/draw_sample.py`; all 41 weeks are listed in `docs/milestone1/sample.csv`. Extreme short = most negative decile of leveraged-fund net position / open interest (−0.376).

Stratum	Weeks drawn
S1 net long (pool 362)	2006-09-26, 2013-01-08, 2021-05-18 … *(+7; full list in repo)*
S2 mod. short pre-2015 (176)	2008-07-22, 2010-05-25, 2014-11-25 … *(+7)*
S3 mod. short 2015+ (449)	2016-06-14, 2018-09-18, 2024-01-02 … *(+7)*
S4 extreme short 2015+ (70)	2024-08-13, 2025-08-26, 2026-07-14 … *(+7)*

B. Pilot results — run 18 September 2026

Week ending 2 September 2025. Ground truth: leveraged funds net −2,640,264 = **−48.5% of open interest**; gross short = **59.88%** of open interest. Both runs used Temporary Chat in **Unpersonalized** mode, one fresh chat each, so memory and custom instructions could not contaminate the baseline. Prompt: positioning table, then "What does this positioning suggest about the Treasury market?" — no mention of basis trades.

Frontier model. Raised the basis trade unprompted; all figures exact.

> *hedge funds frequently short Treasury futures as one leg of the cash-futures basis trade […] Such a position is primarily a relative-value/convergence trade,* ***not necessarily a bet that 10-year yields will rise.*** *[…] So I would* ***not*** *conclude from the CFTC numbers alone that "hedge funds expect Treasury yields to soar."*

Free-tier model. Also raised the basis trade unprompted, but:

> *While some of this is directional macro speculation,* ***the bulk reflects execution of the Treasury Basis Trade*** *[…] Shorts represent* ***over 60% of total open interest***

Measure	Frontier	Free tier
Net positions, OI shares	Correct	Correct
"Over 60% of open interest"	said "about 60%"	59.88% — overstated
Basis trade raised unprompted	Yes	Yes
Asset-manager longs qualified	Yes — "not every long represents a forecast"	No — "overwhelmingly bullish on duration"
Mechanism → magnitude boundary	Hedged (severity 1)	Crossed (severity 2)

What the pilot established. The naive hypothesis — that models read a large short as bearish — is **false at both tiers**. Both models know the basis trade and volunteer it. The failure is narrower and more interesting: the free model

asserted a *proportion* its input cannot yield, applied the position≠view principle to hedge funds but not to asset managers, and overstated a figure it had the data to compute exactly. That is the behaviour §4 is built to measure, and it is why the primary outcome codes *which boundary was crossed* rather than whether the answer was right.

C. AI-use log — first entries

Tool + purpose	Request	Used / rejected	Verification + effect
ChatGPT — research question	Develop a positioning question	Rejected	Gave a two-stage ML forecasting design with no generative-AI system; failed the core requirement
ChatGPT — object of study	Successive critiques of our design	Used	Identified a confound in a single tool-augmented arm, and that a design assuming quantitative grounding helps is weaker than one testing it
ChatGPT — free tier	Pilot arm A	Used as data	Its own output is finding B above
Claude Code — pipeline, pilots, drafting	Ingest, run pilots, draft	Used, under review	API reconciled vs official CFTC archives (2,310 cells, 0 mismatches); 58 tests; group reruns pilots

D. Decision log — first entries

Date	Decision	Reason	Owner
[date]	Filter CFTC by contract code, never name	10Y renamed Feb 2022; name filter silently returns 817 or 240 rows, not 1,057	*[name]*
[date]	Do not assume Tuesday snapshots	14 of 1,057 fall on Mon/Wed in holiday weeks; crashed our first implementation	*[name]*
[date]	Reframe from forecasting to studying the model	Original design had no generative-AI system	*[name]*
[date]	Abandon the "models misread positioning" hypothesis	Pilot falsified it at both model tiers before any build work	*[name]*
[date]	Primary outcome = which boundary crossed, not error rate	Both models were factually right and epistemically different	*[name]*
[date]	Drop the agentic arm; split C into bare/annotated	Agency is an engineering question; labelling the residual is the testable one	*[name]*

Reflexive note. The causalisation error §4 tests for is one this group made: an earlier design of ours treated the model residual as a signal about trader intent. That the error is natural rather than contrived is itself part of the evidence.